

Name:

INTRODUCTION TO CRYPTOGRAPHY & SECURITY

Student ID:

Exam duration: 60 minutes.

## Multiple-Choice Questions

1. What is the most appropriate cryptographic primitive to use if a company wants to distribute authenticated software updates to its customers?
  - A. Hash function.
  - B. Digital signature scheme.
  - C. Message authentication code.
  - D. Block cipher
  - E. Public-key encryption scheme.
2. What is the most appropriate cryptographic primitive to use if an individual wants to ensure confidentiality of the files stored on her hard drive?
  - A. Message authentication code.
  - B. Block cipher.
  - C. Public-key encryption.
  - D. Hash function.
  - E. Private-key encryption.
3. Which of the following best characterizes Discretionary Access Control (DAC)?
  - A. Access decisions based on security labels.
  - B. An entity can grant access to others.
  - C. Centralized control by a security administrator.
  - D. Access based on roles within an organization.
4. Which of the following statements is true about the basic Diffie-Hellman key-exchange protocol.
  - A. The basic protocol provides key exchange secure against active adversaries that can inject and modify messages.
  - B. As with RSA, the protocol only provides eavesdropping security in the group  $\mathbb{Z}_N^*$  where  $N$  is an RSA modulus.
  - C. The protocol can be converted to a public-key encryption system called the ElGamal public-key system.

5. Let  $H : M \rightarrow T$  be a collision resistant hash function where  $|T|$  is smaller than  $|M|$ . Which of the following properties is implied by collision resistance?
  - A. Given  $H(m)$  the attacker learns nothing about  $m$ .
  - B. It is difficult to find  $m_0$  and  $m_1$  such that  $H(m_0) = H(m_1) + 1$ . (here we treat the outputs of  $H$  as integers)
  - C. For all  $m$  in  $M$ ,  $H(m)$  must be shorter than  $m$ .
  - D. It is difficult to construct two distinct messages  $m_0, m_1$  such that  $H(m_0) = H(m_1)$ .
6. In the Needham–Schroeder protocol, why does Alice send a nonce  $r_A$  to the KDC?
  - A. To serve as the session key.
  - B. To ensure that the response from the KDC is fresh and not a replay.
  - C. To encrypt her request.
  - D. To prove that she knows Bob’s identity.
7. What is a digital certificate?
  - A. A password for the root user.
  - B. An encrypted email.
  - C. A document binding a user’s identity to their public key, signed by a Certificate Authority (CA).
  - D. A list of revoked keys.
8. How does a user verify a digital certificate?
  - A. By calling the owner of the certificate.
  - B. By checking the expiration date only.
  - C. By using the trusted public key of the Certificate Authority (CA) to verify the CA’s digital signature on the certificate.
  - D. By decrypting the certificate with their own private key.
9. How does a Man-in-the-Middle attacker exploit the Diffie–Hellman key exchange?
  - A. By guessing the private exponents  $a$  and  $b$ .
  - B. By intercepting the exchanged public values and sending their own public values to each party.
  - C. By shutting down the network.
  - D. By forcing the users to use a small prime number  $p$ .

10. An email authentication system is designed to allow users to log in remotely and to protect against replay attacks.

Indicate which of the following designs provides the **best security**, while allowing a user Alice to set up her account for exactly **one year** (with no updates), assuming Alice checks her email about three times on a normal weekday and about once a day on weekends.

- A. Applying the S/Key Lamport scheme using a hash chain of size 1000.
  - B. The user password is hashed together with a timestamp (rounded to the hour), and the result is sent to the server for authentication.
  - C. Applying a challenge–response mechanism where the challenge is a random number of 10 bits.
  - D. Applying a challenge–response mechanism where the challenge is the number of seconds since the last hour mark according to the server’s clock (e.g., if the current time is 10:05:07 then the challenge is  $307 = 60 \times 5 + 7$ ).
11. Consider the following authentication protocol between Alice and Bob:
- 1. Bob  $\rightarrow$  Alice:  $n_B$
  - 2. Alice  $\rightarrow$  Bob:  $n_A \parallel \text{sig}_A(n_A \parallel n_B \parallel ID_B)$
  - 3. Bob  $\rightarrow$  Alice:  $n'_B \parallel \text{sig}_B(n'_B \parallel n_A \parallel ID_A)$

What is the primary objective of Alice’s message in step 2?

- A. To send her nonce  $n_A$  to Bob.
  - B. To prove her identity to Bob and acknowledge Bob’s nonce  $n_B$ .
  - C. To establish a shared secret key with Bob.
  - D. To request Bob’s public key.
12. Let  $X$  be a uniform random variable over the set  $\{0, 1\}^n$ . Let  $Y$  be an arbitrary random variable over the set  $\{0, 1\}^n$  (not necessarily uniform) that is independent of  $X$ . Define the random variable  $Z = X \oplus Y$ . What is the probability that  $Z$  equals  $0^n$ ?
- A. 0.5
  - B.  $1/2^n$
  - C.  $2/2^n$
  - D. 0

13. Which of the following numbers is a generator of  $\mathbb{Z}_{17}^*$ ?
- A. 2,  $\langle 2 \rangle = \{1, 2, 4, 8, 16, 15, 13, 9, 1, 2, 4, 8, 16, 15, 13, 9\}$
  - B. 3,  $\langle 3 \rangle = \{1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6\}$
  - C. 5,  $\langle 5 \rangle = \{1, 5, 8, 6, 13, 14, 2, 10, 16, 12, 9, 11, 4, 3, 15, 7\}$
  - D. 7,  $\langle 7 \rangle = \{1, 7, 15, 3, 4, 11, 9, 12, 16, 10, 2, 14, 13, 6, 8, 5\}$
  - E. 6,  $\langle 6 \rangle = \{1, 6, 2, 12, 4, 7, 8, 14, 16, 11, 15, 5, 13, 10, 9, 3\}$
  - F. 8,  $\langle 8 \rangle = \{1, 8, 13, 2, 16, 9, 4, 15, 1, 8, 13, 2, 16, 9, 4, 15\}$
  - G. 9,  $\langle 9 \rangle = \{1, 9, 13, 15, 16, 8, 4, 2, 1, 9, 13, 15, 16, 8, 4, 2\}$
  - H. 10,  $\langle 10 \rangle = \{1, 10, 15, 14, 4, 6, 9, 5, 16, 7, 2, 3, 13, 11, 8, 12\}$
14. Let  $G$  be a finite cyclic group of order  $q$  and  $g$  is a generator of  $G$ . Suppose that the discrete log problem is hard in  $G$ . Which of the following problems is also hard:
- A. Randomly pick  $y \in G$ , find  $x$  such that  $g^x = y$ .
  - B. Randomly pick two values  $x \in \mathbb{Z}_q$  and  $y \in G$ , compute  $y^x \cdot g$ .
  - C. Randomly pick  $x \in \mathbb{Z}_q$ , find  $y$  such that  $g^x = y$
  - D. Find  $x$  and  $y$  such that  $g^x = y$ .
  - E. Randomly pick  $y \in G$ , find  $x$  such that  $g^{x+1} = y$ .
15. An access control system according to the RBAC<sub>1</sub> model has four system files  $A, B, C$ , and  $D$ , and four roles denoted by  $P, Q, R$ , and  $S$ .
- In the mathematical representation of this model, the *Permission Assignment (PA)* relation is defined as a set of triplets describing permissions as follows:
- $(P, A, RW), (P, B, RX), (P, C, RWX), (Q, A, RX), (Q, B, O), (Q, C, RX), (Q, D, O),$   
 $(R, A, RWX), (R, B, ORX), (R, C, RWX), (R, D, O), (S, A, R), (S, B, RX), (S, C, RX)$

In the initial phase of this system, there are four users with the following roles:

- Alice has role  $R$ ,
- Bob has role  $P$ ,
- Cathy has role  $Q$ ,
- Dave has role  $S$ .

Which of the following statements are likely to be true?

- A. Alice is Bob's boss but not Cathy's.
- B. Bob is not working under Cathy's supervision but under Alice's.
- C. Alice has Read, Write, and eXecute permissions on file  $A$ .
- D. File  $D$  has only one owner.
- E. Bob is no one's boss.

16. Let  $(S, V)$  be a secure MAC defined over  $(K, M, T)$  where  $M = \{0, 1\}^n$  and  $T = \{0, 1\}^{128}$  (i.e. the key space is  $K$ , message space is  $\{0, 1\}^n$ , and tag space is  $\{0, 1\}^{128}$ ). Which of the following is a secure MAC: (as usual, we use  $\parallel$  to denote string concatenation).

A.

$$\begin{aligned} S'(k, m) &= S(k, m[0, \dots, n-2] \parallel 0) \\ V'(k, m, t) &= V(k, m[0, \dots, n-2] \parallel 0, t) \end{aligned}$$

B.

$$\begin{aligned} S'(k, m) &= S(k, m)[0, \dots, 126] \\ V'(k, m, t) &= [V(k, m, t \parallel 0) \text{ or } V(k, m, t \parallel 1)] \end{aligned}$$

(i.e.,  $V'(k, m, t)$  outputs “1” if either  $t \parallel 0$  or  $t \parallel 1$  is a valid tag for  $m$ ).

C.

$$\begin{aligned} S'(k, m) &= S(k, m \parallel m) \\ V'(k, m, t) &= V(k, m \parallel m, t). \end{aligned}$$

D.

$$\begin{aligned} S'(k, m) &= S(k, m) \\ V'(k, m, t) &= \begin{cases} V(k, m, t) & \text{if } m \neq 0^n \\ \text{“1”} & \text{otherwise} \end{cases} \end{aligned}$$

E.

$$\begin{aligned} S'(k, m) &= S(k, m) \quad \text{and} \\ V'(k, m, t) &= [V(k, m, t) \text{ or } V(k, m \oplus 1^n, t)] \end{aligned}$$

(i.e.,  $V'(k, m, t)$  outputs “1” if  $t$  is a valid tag for either  $m$  or  $m \oplus 1^n$ ).

F.

$$\begin{aligned} S'((k_1, k_2), m) &= (S(k_1, m), S(k_2, m)) \quad \text{and} \\ V'((k_1, k_2), m, (t_1, t_2)) &= [V(k_1, m, t_1) \text{ and } V(k_2, m, t_2)] \end{aligned}$$

(i.e.,  $V'((k_1, k_2), m, (t_1, t_2))$  outputs “1” if both  $t_1$  and  $t_2$  are valid tags).

17. Suppose a MAC system  $(S, V)$  is used to protect files in a file system by appending a MAC tag to each file. The MAC signing algorithm  $S$  is applied to the file contents and nothing else. What tampering attacks are not prevented by this system?

A. Swapping two files in the file system.

- B. Replacing the tag and contents of one file with the tag and contents of a file from another computer protected by the same MAC system, but a different key.
  - C. Replacing the contents of a file with the concatenation of two files on the file system.
  - D. Erasing the last byte of the file contents.
18. Given an RSA signature scheme with the public key  $(n = 9797, e = 7)$ , which of the following signatures are valid?
- A.  $(m = 2601, \sigma = 7496)$
  - B.  $(m = 1894, \sigma = 8667)$
  - C.  $(m = 4707, \sigma = 4976)$
  - D.  $(m = 2016, \sigma = 8159)$

## Calculation Questions

19. Suppose you are told that the one time pad encryption of the message “gui duc 100d” is

5c73d5340a948c86981bc294

(the plaintext letters are encoded as 8-bit ASCII and the given ciphertext is written in hex). The one time pad encryption of the message “gui duc 211d” under the same OTP key is \_\_\_\_\_.

*Note:* you just write the 8-bit ASCII code of the ciphertext in hexadecimal.

.....  
 .....

20. Using Table 1, the inverse of the bytes 29 is \_\_\_\_\_.

.....  
 .....

| X | Y  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|   | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
| 0 | 00 | 01 | 8D | F6 | CB | 52 | 7B | D1 | E8 | 4F | 29 | C0 | B0 | E1 | E5 | C7 |
| 1 | 74 | B4 | AA | 4B | 99 | 2B | 60 | 5F | 58 | 3F | FD | CC | FF | 40 | EE | B2 |
| 2 | 3A | 6E | 5A | F1 | 55 | 4D | A8 | C9 | C1 | 0A | 98 | 15 | 30 | 44 | A2 | C2 |
| 3 | 2C | 45 | 92 | 6C | F3 | 39 | 66 | 42 | F2 | 35 | 20 | 6F | 77 | BB | 59 | 19 |
| 4 | 1D | FE | 37 | 67 | 2D | 31 | F5 | 69 | A7 | 64 | AB | 13 | 54 | 25 | E9 | 09 |
| 5 | ED | 5C | 05 | CA | 4C | 24 | 87 | BF | 18 | 3E | 22 | F0 | 51 | EC | 61 | 17 |
| 6 | 16 | 5E | AF | D3 | 49 | A6 | 36 | 43 | F4 | 47 | 91 | DF | 33 | 93 | 21 | 3B |
| 7 | 79 | B7 | 97 | 85 | 10 | B5 | BA | 3C | B6 | 70 | D0 | 06 | A1 | FA | 81 | 82 |
| 8 | 83 | 7E | 7F | 80 | 96 | 73 | BE | 56 | 9B | 9E | 95 | D9 | F7 | 02 | B9 | A4 |
| 9 | DE | 6A | 32 | 6D | D8 | 8A | 84 | 72 | 2A | 14 | 9F | 88 | F9 | DC | 89 | 9A |
| A | FB | 7C | 2E | C3 | 8F | B8 | 65 | 48 | 26 | C8 | 12 | 4A | CE | E7 | D2 | 62 |
| B | 0C | E0 | 1F | EF | 11 | 75 | 78 | 71 | A5 | 8E | 76 | 3D | BD | BC | 86 | 57 |
| C | 0B | 28 | 2F | A3 | DA | D4 | E4 | 0F | A9 | 27 | 53 | 04 | 1B | FC | AC | E6 |
| D | 7A | 07 | AE | 63 | C5 | DB | E2 | EA | 94 | 8B | C4 | D5 | 9D | F8 | 90 | 6B |
| E | B1 | 0D | D6 | EB | C6 | 0E | CF | AD | 08 | 4E | D7 | E3 | 5D | 50 | 1E | B3 |
| F | 5B | 23 | 38 | 34 | 68 | 46 | 03 | 8C | DD | 9C | 7D | A0 | CD | 1A | 41 | 1C |

Table 1: Multiplicative Inverses in  $\text{GF}(2^8)$  for AES S-box

21. The BkExam is a block cipher optimized for use on final exam. It operates on the 26 letters of the English alphabet:

|          | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|----------|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| $m$      | A | B | C | D | E | F | G | H | I | J | K  | L  | M  | N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
| $E_K(m)$ | P | K | X | C | Y | W | R | S | E | J | U  | D  | G  | O  | Z  | A  | T  | N  | M  | V  | F  | H  | L  | I  | B  | Q  |

As the  $\oplus$  operation is not defined on the set  $\{A, \dots, Z\}$ , we replace it here during encryption with modulo-26 addition (e.g.,  $C \oplus D = F$  and  $Y \oplus C = A$ ).

The ciphertext of the message: “DHBK” which were encrypted using **CBC mode** with  $\text{IV}=H$  is \_\_\_\_\_.

*Note:* You should add IV to the ciphertext.

.....

.....

22. Let  $m$  be a message consisting of  $\ell$  AES blocks (say  $\ell = 100$ ). Alice encrypts  $m$  using randomized counter mode and transmits the resulting ciphertext to Bob. Due to a network error, ciphertext block number  $\ell/2$  is corrupted during transmission. All

other ciphertext blocks are transmitted and received correctly. Once Bob decrypts the received ciphertext, how many plaintext blocks will be corrupted? \_\_\_\_\_

.....  
 .....

23. Compute  $30^{-1} \bmod 113$ .

The number  $30^{-1} \bmod 113$  is \_\_\_\_\_.

.....  
 .....

24. Compute  $779^{168002} \bmod 11413$  knowing that  $11413 = (101 \times 113)$ .

The number  $779^{168002} \bmod 11413$  is \_\_\_\_\_.

.....  
 .....

25. What is the discrete log of 4 base 3 in  $\mathbb{Z}_{17}^*$ .

The number  $\text{Dlog}_3(4)$  is \_\_\_\_\_.

.....  
 .....

26. An RSA encryption scheme has the set-up parameters  $p = 13$  and  $q = 19$ . The public key is  $e = 17$ . You are asked to decrypt the ciphertext  $y = 16$ .

The plaintext  $x$  is \_\_\_\_\_.

.....  
 .....

27. Let  $\mathbb{Z}_{19}^*$  be a cyclic group and 3 is a generator. Compute  $\text{DH}_3(12, 15)$  in this group.  
The value  $\text{DH}_3(12, 15)$  is \_\_\_\_\_.

.....  
.....

28. Given an elliptic curve

$$E : \quad y^2 = x^3 + 2x + 2 \pmod{17}$$

The base point  $P = (13, 10)$ . Compute the point  $Q = 78 \cdot P$ .

The point  $Q$  is \_\_\_\_\_.

*Note:* Write the result as  $(\mathbf{p}, \mathbf{q})$  with no space in between.

For the sake of calculation, the points that are multiples of the primitive element  $(5, 1)$  are listed in the Table below.

| $k$         | 1       | 2      | 3      | 4     | 5      | 6       | 7      | 8      | 9             | 10     |
|-------------|---------|--------|--------|-------|--------|---------|--------|--------|---------------|--------|
| $k \cdot G$ | (5,1)   | (6,3)  | (10,6) | (3,1) | (9,16) | (16,13) | (0,6)  | (13,7) | (7,6)         | (7,11) |
| $k$         | 11      | 12     | 13     | 14    | 15     | 16      | 17     | 18     | 19            |        |
| $k \cdot G$ | (13,10) | (0,11) | (16,4) | (9,1) | (3,16) | (10,11) | (6,14) | (5,16) | $\mathcal{O}$ |        |

.....  
.....

29. Given an elliptic curve

$$E : \quad y^2 = x^3 + 2x + 2 \pmod{17}$$

Consider the point  $P = (13, 10)$ . Alice and Bob are to compute a shared key in a Diffie-Hellman protocol based on elliptic curve  $E$ . That is,

- Alice randomly pick a number  $a$  and compute the point  $aP = (16, 13)$ , and send to Bob;
- Alice receive the point  $bP = (16, 13)$  from Bob.

The shared key  $abP$  between Alice and Bob is \_\_\_\_\_.

*Note:* Write the result as  $(\mathbf{p}, \mathbf{q})$  with no space in between.

For the sake of calculation, the points that are multiples of the primitive element  $(5, 1)$  are listed in the Table below.

|             |         |        |        |       |        |         |        |        |               |        |
|-------------|---------|--------|--------|-------|--------|---------|--------|--------|---------------|--------|
| $k$         | 1       | 2      | 3      | 4     | 5      | 6       | 7      | 8      | 9             | 10     |
| $k \cdot G$ | (5,1)   | (6,3)  | (10,6) | (3,1) | (9,16) | (16,13) | (0,6)  | (13,7) | (7,6)         | (7,11) |
| $k$         | 11      | 12     | 13     | 14    | 15     | 16      | 17     | 18     | 19            |        |
| $k \cdot G$ | (13,10) | (0,11) | (16,4) | (9,1) | (3,16) | (10,11) | (6,14) | (5,16) | $\mathcal{O}$ |        |

.....

.....

30. We consider the ElGamal signature scheme. You are given Bob's private key  $sk = d = (12)$  and the corresponding public key  $pk = (p, g, g^d) = (29, 2, 7)$ . Compute the Elgamal signature  $(r, s)$  for a message from Bob to Alice with message  $m = 20$  and ephemeral key  $k_E = 5$ .

The signature  $(r, s)$  is \_\_\_\_\_.

*Note:* Write the signature as  $(\mathbf{r}, \mathbf{s})$  with no spaces in between.

.....

.....